

ASSIGNMENT - 2

Course name - Database Management System

Course code - ENCS205

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Section - A

Section-A

1) Fill in the blanks:

- 1) The Entity Relationship (ER) model visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.
- 2) In the Relational Model, data is organized into tables, with relationship b/w entities defined through foreign key constraints.
- 3) The Object-Oriented model treats data as collection of objects, each with unique attributes & methods, mirroring real-world entities & interactions.
- 4) Integrity Constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

Section-B (Case Study based)

- 2) In a healthcare system, discuss how data abstraction can simplify the interaction b/w healthcare professional & patient records.

Ans:- 1) Data abstraction is an important concept in database systems that hides unnecessary details & only shows the required information to the users. In a healthcare system, this helps doctors, nurses

and help staff access patient information easily without dealing with the complex structure of the database.

- 2) For example, a doctor only needs to see patient details such as name, age, medical history, test reports & prescribed medicines. The doctor does not need to know how the data is stored in the database tables. Data abstraction hides these technical details & presents the information in a simple form.
- 3) Similarly, a receptionist may only need basic information such as patient name, appointment time, & doctor details. They do not need access to medical records or treatment history. Data abstraction allows different users to see only the information relevant to their role.
- 4) Therefore, data abstraction makes healthcare systems easier to use, improves efficiency & helps healthcare professionals quickly access important patient information.
- 5) Explore the potential benefits & challenges of implementing data abstraction in healthcare system.

Ans: # Benefits:-

1) Improved Data Security:-

- Sensitive patient information is protected.
- Only authorized users can access medical records.

2) Ease of Use:-

- Healthcare staff can easily access the required information without understanding database understanding.

aster Decision Making :-

→ Doctor can quickly view patient history & provide treatment.

4) Better Data organization :-

→ Patient records are stored in a systematic & structured format.

5) Flexibility in System Design :-

→ If the database structure changes, users are not affected because they only see the abstract view.

Challenges :-

1) Complex System Design :-

→ Designing proper abstraction level requires careful planning.

2) Development Time :-

→ Creating different user views may increase development time.

3) Security Management :-

→ Developers must ensure that unauthorized users cannot access restricted data.

4) Maintenance Effort :-

→ The system requires regular updates & monitoring.

Section - C (Simulation Based)

4) Design a database for a University using the Entity-Relationship (ER) model.

Ans- Main Entities :- 1) Student, 2) Course, 3) Professor

→ Attributes :-

- 1) Student
 - Student_ID (Primary Key)
 - Student_Name
 - Age
 - Gender
 - Email
- Department

2) Course Entity :-

- Course-ID (Primary key)
- Course-Name
- Credits
- Department

3) Professor Entity

- Professor-ID (Primary key)
- Professor-Name
- Department
- Email

→ Relationships b/w Entities :-

1) Student - Course Relationship (Enrollment)

- A student can enroll in multiple courses.
- A course can have multiple students.
- This is a Many-to-Many relationship.

→ Professor - Course Relationship (Teaching)

- A professor can teach multiple courses.
- Each course is usually taught by one professor.
- This is One-to-many relationship.

ER Diagram :-

